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Mesh Convergence Report

In this study, we performed a mesh convergence test on a cantilever beam by systematically refining the mesh and monitoring the resulting beam deflection at the central node on the free end. The mesh density was incrementally increased while maintaining cubic elements to preserve shape consistency.

Two simulations were performed using mesh sizes of 0.009 m and 0.005 m. The finer mesh (0.005 m) consisted of 60 total elements and resulted in a beam deflection of $-2.1245*10^{-7}$ m. The coarser mesh (0.009 m) had 36 elements and produced a deflection of $-2.1173*10^{-7}$ m. Using the finer mesh deflection as the reference, we calculated the percent error for the coarser mesh as 0.339%.

$$Percent\ Error\ = \frac{LastMeshBeamDeflection - CurrentMeshBeamDeflection}{LastMeshBeamDeflection} *\ 100$$

$$Percent\ Error\ = \frac{-2.1245\times10^{-7} - 2.1173\times10^{-7}}{-2.1245\times10^{-7}} *\ 100\ =\ 0.339\%$$

Despite a slight increase in computational time (0.938 s → 1.047 s), the deflection results indicate good convergence. The change in deflection is minimal, suggesting that further mesh refinement would result in diminishing returns while increasing computation time.

The mesh with 60 total elements and cubic geometry provides a good balance between accuracy and computational cost for this simulation. It captures the structural behavior of the beam with under 0.5% error compared to the finer mesh, making it a suitable choice for future simulations.

| **Mesh Size (m)** | **Total Elements** | **Beam Deflection (m)** | **Percent Error (%)** | **CP Time (s)** |
|---|---|---|---|---|
| 0.005 | 60 | $-2.1245*10^{-7}$ | 0.000 | 0.938 |
| 0.009 | 36 | $-2.1173*10^{-7}$ | 0.339 | 1.047 |

**Table 1:** Final Mesh Convergence Table

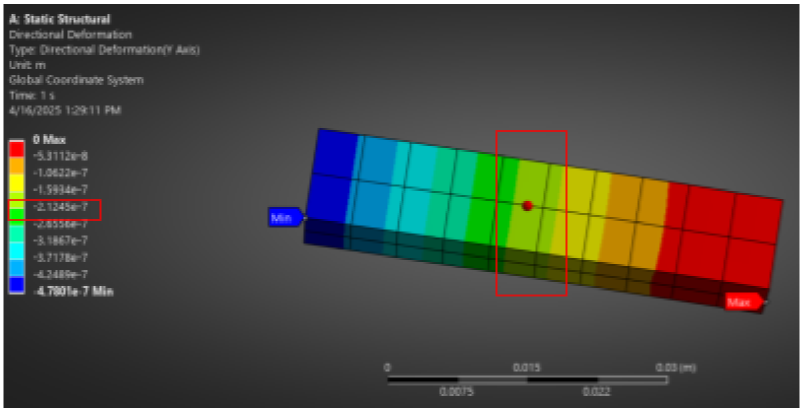


**Fig 1:** Finer mesh (0.005 m) consisted of 60 total elements and resulted in a beam deflection of $-2.1245*10^{-7}$ m.

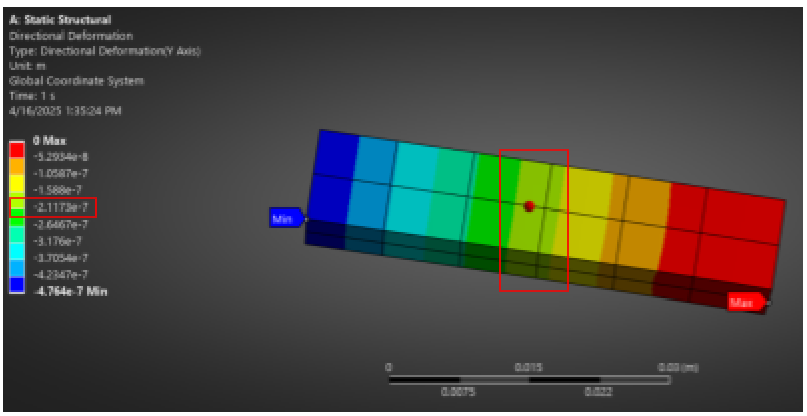


**Fig 2:** Coarser mesh (0.009 m) had 36 elements and produced a deflection of $-2.1173*10^{-7}$ m.

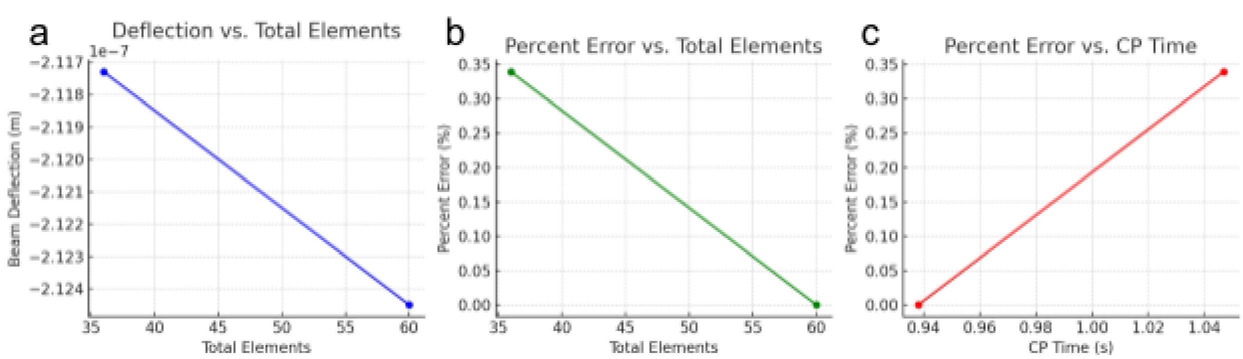


**Fig 3: a)** Deflection vs. Total Elements shows that as the mesh is refined, the deflection converges to a stable value. **b)** Percent Error vs. Total Elements highlights that the coarser mesh introduces a ~0.34% error. **c)** Percent Error vs. CP Time illustrates the tradeoff between computational cost and accuracy.